Predictive Analytics Poster Rubric (Poster-Only Assessment)

#	Criterion	Points	Exemplary (Full Points)	Competent (Partial Points)	Needs Improvement (Minimal Points)
1	Prediction Goal(s)	5	5 pts: Clear, specific predictive objective(s); response type (regression/classification) explicitly identified and justified; names the stakeholder and the decision the predictions support; strong business/research rationale.	3–4 pts: Goal and response type stated, but the stakeholder, the decision, or the rationale is thin; limited link to context.	0–2 pts: Goal unclear/missing; response type unspecified; no link to a stakeholder's decision.
2	Dataset Overview	8	7–8 pts: Reports #obs/#features; types for key features; response variable identified/typed; data provenance/window stated.	4–6 pts: Size and response identified; partial feature typing; minimal provenance.	0–3 pts: Missing size/response ID; little/no feature typing or context.
3	Exploratory Data Analysis (EDA)	12	10–12 pts: Appropriate summaries (response + key predictors); ≥2 relevant plots; interprets skew/outliers/imbalance/seasonality; connects insights to modeling choices and metric selection.	6–9 pts: Basic stats and ≥1 plot; some interpretation but limited linkage to modeling.	0–5 pts: Incomplete stats; weak/irrelevant visuals; minimal/no interpretation.
4	Data Preparation (Feature Engineering, Missing Data, etc.)	18	15–18 pts: ≥1 non-trivial new feature (interactions, lags/rolling stats, domain-driven ratios, text-derived signals) with clear rationale; quantifies missingness by feature; justified strategy (drop/impute/advanced) with leakage-safe implementation; demonstrates impact via ablation/CV deltas or sensitivity to imputation choices.	9–14 pts: New feature with partial rationale; identifies missingness and applies a reasonable method; limited evidence of impact or sensitivity analysis.	0–8 pts: No new/trivial feature; missingness unreported or mishandled; unjustified/inappropriate method; no evidence of value.
5	Baseline Model (Choice, CV, Interpretation)	17	14–17 pts: Appropriate simple baseline (e.g., Linear/Logistic Regression), the incumbent any complex model must beat; pipeline is leakage-safe; k=5/10 CV (time-respecting for forecasting) with a primary metric chosen for the cost of errors; reports the fold scores with their mean and SD (no per-model confidence interval needed); summarizes implications and limitations; proposes concrete next steps.	8–13 pts: Reasonable model; CV present with partial integration/details; metric adequate; interpretation limited or generic.	0–7 pts: Inappropriate/unspecified baseline; no/incorrect CV; unclear metric; raw numbers only; no reflection.

#	Criterion	Points	Exemplary (Full Points)	Competent (Partial Points)	Needs Improvement (Minimal Points)
6	Model Comparison and Champion Selection (Choice, Tuning, Paired Comparison, Rationale)	25	22–25 pts: Model aligned to data/goal (e.g., RF/GBM/SVM/NN); tuning inside CV with leakage controls. Before comparing: primary metric, margin δ (smallest gain that would change the stakeholder's decision, with its business meaning), and candidate order (simplest first). Identical folds; per-fold differences (positive favors the challenger); the verdict from a rule applied correctly: notebook 08's rule (Fall 2026 note) or, recommended, the interval corrected for fold overlap, mean ± t × SD(differences) × √(1/k + 1/(k−1)), named as worthwhile gain (entirely above δ), worthwhile loss (entirely below −δ), practically equivalent (entirely inside ±δ), or unresolved. One champion (the baseline can win); extra complexity only on a worthwhile gain; rationale in the stakeholder's terms (what the gain means, secondary checks, runner-up, what would change the decision). No model's own interval, interval overlap, or lower bound picks the champion; the test set is not used to select.	13–21 pts: Plausible model and tuning, but the comparison or decision is incomplete, or misapplies the rule the team used: folds not paired; δ missing, without business meaning, or set after the results without saying so; verdict from means alone, interval overlap, or a model's own interval; with the corrected interval, unresolved called equivalent; or a champion without a rationale tied to the decision.	0–12 pts: Misaligned/unspecified model; superficial/absent tuning; incorrect CV; no clear comparison and no justified champion.
7	Report Quality & Visual Communication	10	9–10 pts: Clear, scannable poster; logical flow (Intro → Methods → Results → Conclusions); the champion and its main reason are easy to find; legible, labeled figures/tables (titles, axes, units, legends); consistent typography/layout; code refs/appendix where appropriate; minimal grammar issues.	6–8 pts: Generally readable; minor layout/labeling gaps; some figures under-explained or styling inconsistencies.	0–5 pts: Disorganized; illegible/unlabeled visuals; weak linkage between text and graphics; notable grammar/formatting issues.
8	Sources, Acknowledgments, AI Use & Data Privacy	5	5 pts: Data source cited (provider, URL or DOI, version or access date); references support the business-problem claims and the main software is credited (e.g., scikit-learn, pandas); Acknowledgments section names the Mitch Daniels School of Business, Prof. Moreira, the TAs, and other contributors; AI Use Disclosure names each AI tool and what it was used for and states that the team verified the output (or states that no AI tools were used); only aggregated or anonymized data and examples.	3–4 pts: AI Use Disclosure present but vague (names tools without saying what they did), or one other element missing or incomplete (e.g., acknowledgments omit the TAs; data source cited without references or tools). No privacy issue.	0–2 pts: No AI Use Disclosure, or two or more other elements missing. Any personally identifiable or NDA-restricted data on the poster scores 0 here and must be removed before printing.

Total: 100 points. This rubric grades M09 (poster first draft), M10 (final poster), and the Instructor/TA evaluation after the conference.

Criterion 6 at M09 (due Nov 1): the comparison and the champion get feedback, not a deduction, on the first draft; they are scored from M10, under the Fall 2026 note.

Fall 2026 (this note controls criterion 6): the notebooks still teach notebook 08's three-outcome rule; applying it correctly earns full credit, as does a champion chosen as notebook 14 teaches and explained for the stakeholder. The corrected interval is recommended, not required. If δ or your criteria were set after the M06/M08 results, say so: honest disclosure is not penalized. A baseline champion or an honest “unresolved” verdict can earn full credit. No new model search or test-set opening is required.

AI use: per the syllabus AI Policy, every poster carries an AI Use Disclosure (the template has a section for it). Errors in AI-assisted work are the team’s responsibility.

Honors: a labeled honors section, where present, is graded only under the Honors Research Extension and is not scored under this rubric.